International Journal of Cardiology 93 (2004) 175–179

International Journal of Cardiology

www.elsevier.com/locate/ijcard

Efficacy of folic acid when added to statin therapy in patients with hypercholesterolemia following acute myocardial infarction: a randomised pilot trial

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Received 8 September 2002; received in revised form 1 February 2003; accepted 13 February 2003

Abstract

Background: Folic acid is assumed to have favourable effects on vascular endothelium, directly as well as indirectly through its effect on homocysteine metabolism. However, the clinical value of folic acid in secondary prevention after acute myocardial infarction (MI) has never been tested. Thus, a randomised, open-label, multicentre trial was performed in order to study the effect of folic acid 5 mg o.d. when added to statin therapy on the incidence of recurrent major clinical events up to 1 year post-MI. *Methods:* A total of 283 patients with a total cholesterol > 6.5 mmol/l (251 mg/dl) (mean 7.3 mmol/l) were included. All patients received 40 fluvastatin. In 140 of the 283 patients, folic acid (5 mg o.d.) was instituted at discharge, and the remaining 143 patients served as controls. Other secondary prevention measures for both groups were advocated. The primary endpoint was a composite consisting of all vascular events, including death, recurrent MI, strokes, and unplanned invasive coronary interventions. *Results:* At baseline, the two groups were well-matched for all clinical and demographic parameters. After 1 year of treatment, no difference was noticed in the primary endpoint between the two groups. These endpoints occurred in 43 patients (31%) in the folic acid group, as opposed to 45 patients (31%) in the control group. All separate cardiovascular events were also equally distributed between both groups. Total cholesterol levels decreased to a similar extent in the two groups (to 5.5 and 5.7 mmol/l, in folic acid and control groups, respectively). *Conclusions:* In this medium-size pilot study, folic acid did not demonstrate any beneficial additive effects on cardiovascular mortality or morbidity in post-MI patients with hypercholesterolemia who were treated with statin therapy. Larger trials, possibly targeting at selected populations, must be awaited before definitive conclusions regarding the potentially favourable effects of folic acid supplementation in secondary prevention can be drawn.

Keywords: Folic acid; Secondary prevention; Hypercholesterolemia

1. Introduction

Several lines of investigations have pointed out that folates have salutary effects on endothelial function as a surrogate end point for cardiovascular risks [1–3]. Primarily, it has been presumed that this could be explained by an indirect effect whereby folates reduced homocysteine levels in plasma [4]. In addition, it also has been appreciated that elevated homocysteine is related to the prognosis of patients with atherosclerotic coronary artery disease [6–8]. Recently, however, research suggested that folates may have ameliorating actions on endothelial function independent of its homocysteine lowering effects and thus have direct effects on the endothelium [9]. Potential mechanisms include antioxidant

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doi:10.1016/j.ijcard.2003.02.001

actions, effects on cofactor availability or direct interactions with the enzyme endothelial NO synthase. However, thus far, limited data are available on the effect of folate substitution on hard clinical endpoints [8,10]. One recent investigation reports that folic acid substitution, albeit in combination with pyridoxine and vitamin B12, reduces the rate of restenosis and the need for revascularisation after percutaneous coronary angioplasty [11]. Until now, no reports are available on the effect of folic acid supplementation as a single intervention or in combination with other vitamins, on cardiac mortality and morbidity after acute myocardial infarction (MI), which is the subject of the present study.

2. Methods

2.1. Patients

Patients with a diagnosis of acute MI were screened for inclusion in the trial. They were assumed eligible when total cholesterol value, taken at admission or within 24 h after onset of symptoms, was > 6.5 mmol/l (251 mg/dl). For the confirmation of the qualifying MI, an elevation of the myocardial band of creatinine phosphokinase (CK-MB) at least two times the upper limit of normal was required or CK level at least two times the upper limit of the normal. In addition, eligibility for the study also required one of the following: new or markedly increased chest pain lasting longer than 30 min, or classical ECG changes (a new pathological Q wave of ≥ 0.04 s duration or ≥ 25% of the corresponding R wave amplitude, both in at least two contiguous leads).

Main exclusion criteria were: age < 18 years, use of lipid-lowering agents within the previous 3 months, a high triglyceride level > 4.5 mmol/l, known familial dyslipidaemia, low vitamin B12 level, previously known hyperhomocysteinaemia (total plasma homocysteine > 18 μmol/l) or a known disturbed methionine loading test (total plasmahomocysteine > 47 μmol/l), severe renal failure (serum creatinine >180 μmol/l), known hepatic disease, signs and symptoms of severe heart failure (New York Heart Association class IV), and a scheduled percutaneous coronary intervention (PCI) or coronary artery bypass graft (CABG) operation.

2.2. Study treatment and assessments

As soon as eligibility was confirmed, patients were invited to participate in the study. Additionally, written informed consent was obtained. Patients were randomised to treatment with open-label folic acid 5 mg o.d. or not. Statin treatment (fluvastatin 40 mg o.d.) was initiated simultaneously. Folic acid treatment was commenced at least 1 day prior to hospital discharge but not later than 14 days after the onset of the qualifying myocardial event. Treatment was continued for 1 year. It was left to the discretion of the attending physician to implement standard secondary prophylaxis medication as aspirin, beta-blocking agents, and/or ACE inhibitors. If after discharge, residual ischaemia on exercise test or scintigraphy was noted, the physicians were encouraged to take the appropriate routine diagnostic and therapeutic measures, i.e., PCI or CABG. Clinical events were carefully registered during outpatient visits, which were planned at week 6, 12, 26, 40, and 52. At these visits, compliance to folic acid was monitored by pill counts. Blood samples for lipid analysis were only taken at baseline. During the course of the study, dose levels of fluvastatin could be adapted by the attending physicians. In case of intolerance to fluvastatin, or any other reason, other statin drugs could be initiated. No formal target cholesterol values were delineated.

Common practice in the Netherlands at the time advised to pursue a total cholesterol value of < 5 mmol/l. At 12 months, lipid profiles were measured. Systematic homocysteine measurements during the study were not performed.

The study was conducted in accordance with the Declaration of Helsinki as revised in 1996 in 28 participating medical centres in the Netherlands and in all participating centres the local ethics committees had approved the protocol prior to the study.

2.3. Study endpoints

Clinical events were defined as cardiovascular death (sudden death, fatal recurrent MI, fatal stroke, and other cardiovascular deaths), non-cardiovascular death, recurrent MI, or recurrent ischaemia necessitating hospitalisation or revascularisation (PCI, CABG). MI was defined by a CK elevation at least two times the upper limit of normal and in addition one of the following: On EKG, new pathological Q-waves in at least two contiguous leads or new left bundle branch block or chest pain longer than 30 min. Recurrent

Table 1
Baseline characteristics of 283 patients randomly assigned to folic acid (N = 140) or not (N = 143)

	Folic acid	Control
Age (years)	59	59
Male gender (%)	69	70
Caucasian (%)	99	99
CK elevation at baseline (× ULN)	7	7
Anterior myocardial infarction (%)	49	54
Treatment with		
Fibrinolysis (%)	47	43
PTCA (%)	4	1
Both (%)	1	2
Aspirin (%)	97	99
Beta-blocking agents (%)	93	88
ACE inhibitors (%)	36	39
Diuretics (%)	24	24
Calcium channel blockers (%)	18	14
Mean lipid values		
Total cholesterol (mmol/l)	7.3	7.2
LDL cholesterol (mmol/l)	5.2	5.1
HDL cholesterol (mmol/l)	1.2	1.2
Triglycerides (mmol/l)	2.3	2.1

Table 2
Cardiovascular mortality and morbidity*

	Folic acid	Control
Sudden death	3	5
Fatal MI	2	2
Fatal stroke	1	0
Other cardiovascular death	0	0
Stroke	0	0
Recurrent MI	6	8
Recurrent ischemia	6	8
CABG (not pre-planned)	8	14
PTCA (not pre-planned)	24	23
Any event	43	45

* Per clinical event category, the number of patients with at least one event is presented. No patient double counting for multiple events within a category was performed. Between categories, patients are double counted in case of multiple events, e.g., a patient suffering two recurrent myocardial infarctions followed by a CABG is counted once in both categories.

ischemia necessitating hospitalization was defined as a recurrence of typical or atypical anginal pain at rest or at effort with the appearance of a ST segment change ≥ 0.1 mV and/or T wave inversion in at least 2 of 12 leads. Stroke, fatal or non-fatal, was recorded when unequivocal signs of focal or global neurological deficit with sudden onset was present with a duration greater than 24 h and which were judged to be vascular in origin. For fatal cerebrovascular disease to be recorded, these criteria were to be fulfilled and/or the diagnosis should be stated on the death certificate. An Independent Data and Safety Monitoring Committee adjudicated all major clinical events.

2.4. Statistical analysis

All randomised patients fulfilling the eligibility criteria were included in the analysis of time to a major clinical event. Hereby, intention to treat principle is taken into account. Continuous data are presented as mean and standard deviation (S.D.). The two treatment groups were compared using one-way analysis of variance if normally distributed, or by (Mann–Whitney) Wilcoxon's two-sample test if the distribution was skewed. Categorical data are presented by percentage and count of each category. Treatment comparisons were made using Fisher exact or chi-square tests. Time-to-event analysis was performed using log rank tests, and these data are presented as Kaplan–Meier survival curves. The study was powered for a 50% reduction in clinical events on the basis of existing observational data in populations with coronary artery disease [5,6]. We assumed that the 1-year event rate was about 30%. To have 80% power (5% significance level) to detect a relative risk of 0.5, we included at least 120 patients in each treatment group. Analyses were performed using commercially available software (Statistical Analysis System version 6.12, SAS Institute, Cary, NC).

3. Results

3.1. Baseline

A total of 283 patients with an acute MI were included in this study. Of these patients, 140 patients were randomly assigned to folic acid 5 mg o.d. and 143 patients formed the control group. Baseline characteristics are listed in Table 1. In general, patients are included with large infarct size (CK elevation seven times ULN). Concomitant medication and cholesterol levels are depicted in Table 1. The mean vitamin B12 level was 0.30 nmol/l in both groups.

3.2. Clinical events and other follow-up data

Clinical cardiovascular events were evenly distributed in both treatment arms. In total, 43 patients (31%) in the folic

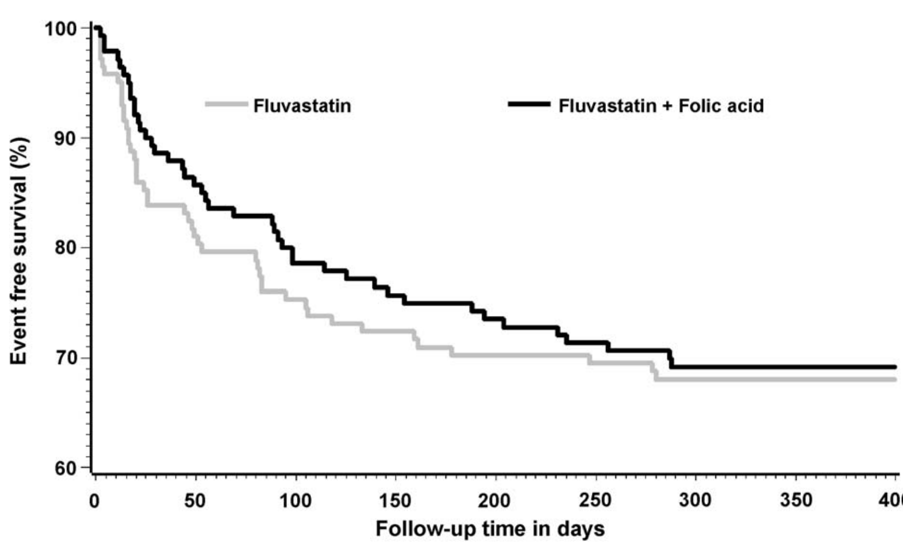

Fig. 1. One year of freedom from cardiovascular mortality and morbidity in patients treated with folic acid versus controls (Kaplan–Meier method). $P = 0.69$.

acid group and 45 patients (31%) in the control group suffered from any major clinical event (Table 2). The time to the first clinical event is depicted in the Kaplan–Meier curve in Fig. 1. There was no statistically significant difference between both groups (log-rank test, $p = 0.69$). Likewise, there is no difference observed in noncardiovascular mortality and morbidity ($p = 0.41$).

Discontinuation of folic acid treatment was noted in 23 patients (premature death, $n = 7$, and intercurrent event, $n = 16$). Average exposure to folic acid was 335 days. In 20% of the cases, fluvastatin was changed to any other statin. This change was evenly distributed in both groups. At 12 months, lipid parameters were similar for both groups; mean total cholesterol value was 5.5 mmol/l (S.D. 0.9) in the folic acid group and 5.7 mmol/l (S.D. 0.9) in the control group. Mean LDL values were 3.3 mmol/l (S.D. 0.8) and 3.4 mmol/l (S.D. 0.8), respectively. No significant changes in mean HDL or triglycerides were observed. In case of the use of fluvastatin, the mean dose at 12 months was 52 mg (S.D. 18) and 53 mg (S.D. 17), respectively.

4. Discussion

In this pilot study, the addition of folic acid to post-MI patients with hypercholesterolemia who were also treated with statin therapy, did not show any beneficial effect on cardiovascular mortality or morbidity in the first year after MI. Although we examined a high-risk population, which is illustrated by the high incidence of recurrent events (31%), the relatively small sample size certainly does not allow definite conclusions. Indeed, a possible favourable effect cannot be excluded when the same study was conducted in a larger population or with a longer follow-up. The present data do, however, confirm the safety of folic acid administration in this population.

One possible explanation for the lack of effect could be that the present population was treated rather aggressively, reflected by a high incidence of PCI and CABG. In addition, the majority of patients was treated with aspirin and beta-blocking agents. Nevertheless, from the perspective of current guidelines, the observed lipid parameters could even be more stringently pursued.

Furthermore, in this study, a 5-mg dose of folic acid is used. This is mainly due the fact that this was the only commercial available dose at the time when the study was designed. At present, a dose as low as 0.5 mg is probably sufficient when the main goal is to pursue lower homocysteine levels [12]. However, it has also been recognized that a dose of 5 mg enhances endothelial function in patients with hypercholesterolaemia, or in patients with coronary heart disease [1,4].

Recently, it has been demonstrated that normalisation of endothelial function in coronary artery disease by folic acid 5 mg is independent of its action on homocysteine [2]. Lower doses, however, have not yet been tested in a comparable way. Additionally, no data are available concerning favourable clinical outcome in terms of hard clinical endpoints. Moreover, in monkeys, folic acid prevents hyperhomocysteinemia, yet, it does not prevent vascular dysfunction or development of atherosclerotic lesions [13].

Currently ongoing, much larger studies use a comparable dose of 5 mg folic acid (CHAOSTWO) or a lower dose of 2 mg in conjunction with low dose of vitamin B12 (SEARCH). In the present study, we deliberately used only folic acid supplementation. As a safety measure, patients with low vitamin B12 level were excluded. It is also noteworthy that in the Netherlands, contrary to the USA and Canada, no fortification of cereal grain flour products with folic acid has yet been instituted.

A limitation of the present study is that we did not measure plasma homocysteine levels in our patients, neither at baseline nor after treatment. The rationale behind this was that the assumed effect, if any, would only partly be dependent on the reduction of plasma homocysteine levels. Nevertheless, these measurements could have been interesting and would have allowed further analysis of the data.

5. Conclusions

In this study, administration of folic acid, when added to statin therapy, did not have any beneficial effects on cardiovascular mortality or morbidity in a population of post-AMI patients with a relative high cholesterol value at admission. Given the sample size of the population (283 patients), results of much larger, ongoing trials must be awaited before definitive conclusions can be drawn.

Acknowledgements

Investigators: A.H. Liem (Goes), A.J. van Boven (Groningen), R.M. Robles de Medina (Den Haag), A.J.A.M. Withagen (Delft), J. Ansink (Lelystad), B.J. van den Berg (Capelle a/d Ijssel), P.J.L.M. Bernink, J. Posma (Groningen), L.H.M. Bouwens (Deventer), J.J.J. Bucx, S.A.M. Saïd (Hengelo), P.H.J.M. Dunselman (Breda), J.G. Engbers (Emmen), M.A Galjee (Enschede), B.J.B. Hamer (Amersfoort), D.P. Hertzberger (Nijmegen), L.J. Jansen (Amsterdam), R. de Kan (Zierikzee), L.H.J. van Kempen (Velp), J.P. Kerker (Rotterdam), P.A.R. de Milliano (Hilversum), A. Oomen (Sneek), M.G. Scheffer (Rotterdam), L.C. Slegers (Veldhoven), S. Strikwerda (Breda), P. Sijbring (Winterswijk), C.M.J. Uitdehaag (Doetinchem), G.J.E. Verdel (Haarlem), H.A. Werner (Schiedam), and P. Zijnen (Tilburg).

Contributors: Liem, van Boven, Withagen, Robles de Medina, and Tijssen were involved in the design of the study; Veeger was responsible for data handling and statistical analysis. Liem, van Boven, Withagen, and Robles de Medina were members of the Steering Committee and responsible for the conduct of the trial in their centres. Tijsen was a member of the Independent Monitoring Committee. Liem wrote the paper. All investigators, including van Veldhuisen, discussed the results, revised the report critically, and approved the final version.

The authors would like to thank Wendy Hendriks and Maribel van der Pas for supervising this multicentre trial and as well as AstraZeneca, The Netherlands, for an unrestricted grant to finance the study. This study was conducted under the auspices of the Working group on Cardiovascular research (WCN), The Netherlands. Dr. Van Veldhuisen is an Established Investigator of the Netherlands Heart Foundation (grant D97.017).

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